

AI Talent Acquisition System - Final Project Structure

This project is organized as four member-owned model components that are combined through shared backend and frontend integration layers.

Run The Project

Backend:

```
cd backend
source .venv/bin/activate
uvicorn app.main:app --reload
```

Frontend:

```
cd ta
npm install
npm run dev
```

URLs:

- ? Backend: <http://127.0.0.1:8000>
- ? Frontend: <http://localhost:5173>
- ? Swagger API docs: <http://127.0.0.1:8000/docs>

Demo logins:

- ? Admin: admin@talentai.local / Admin123!
- ? HR: hr@talentai.local / Recruiter123!
- ? Candidate: candidate@talentai.local / Candidate123!

Top-Level Folders

- ? backend/ - FastAPI backend, model APIs, shared ATS integration, data store, and model artifacts.
- ? ta/ - React + Vite frontend.
- ? models/ - four model training notebooks used as the reference for member ownership.
- ? RUN_GUIDE.md - short run instructions.
- ? MODULE_DIVISION.md - compact four-member ownership map.

Four Model Notebooks

- ? models/01_attrition_risk_model_training.ipynb - Member 1, Attrition Risk.
- ? models/02_resume_parser_model_training.ipynb - Member 2, Resume Parser.
- ? models/03_job_matching_model_training.ipynb - Member 3, Job Matching.
- ? models/04_interview_evaluation_model_training.ipynb - Member 4, Interview Evaluation.

Backend Structure

Backend entrypoint:

- ? backend/app/main.py
- ? Creates the FastAPI app.
- ? Registers CORS.
- ? Loads startup model bundles.
- ? Registers the four member routers.
- ? Registers the shared hiring-cycle router.
- ? Provides /health.

Shared backend layer:

- ? backend/app/shared/ats_store.py - JSON ATS datastore, seeded users, companies, vacancies, applications, interviews, and audit trail.
- ? backend/app/shared/cv_cache.py - temporary CV cache used by resume and attrition flows.
- ? backend/app/shared/errors.py - shared friendly API error helper.

- ? backend/app/shared/hiring_cycle.py - combined ATS workflow: auth, jobs, applications, recruiter review, interviews, decisions, dashboards.
- ? backend/app/shared/hiring_cycle_schemas.py - request/response schemas for ATS workflow.
- ? backend/app/shared/model_loader.py - attrition model artifact loader.
- ? backend/app/shared/schemas.py - schemas for standalone model APIs.
- ? backend/app/shared/utlis.py - shared parsing, file, upload, and feature helpers.

Backend data and scripts:

- ? backend/data/ats_store.json - persistent demo ATS datastore.
- ? backend/uploads/.gitkeep - upload folder placeholder.
- ? backend/scripts/train_resume_explainer_model.py - trains Member 2 resume explainer artifact.
- ? backend/scripts/train_interview_model.py - trains Member 4 interview scorer artifact.
- ? backend/requirements.txt - Python dependencies.

Member 1 - Attrition Risk

Purpose:

- ? Predict early attrition risk.
- ? Return attrition probability, retention probability, risk band, and top factors.

Backend:

- ? backend/app/member_1_attrition_risk/routes.py - attrition API endpoints.
- ? backend/app/member_1_attrition_risk/predictor.py - feature preparation and XGBoost prediction.
- ? backend/app/member_1_attrition_risk/model_artifacts/attrition_xgb.json - XGBoost model.
- ? backend/app/member_1_attrition_risk/model_artifacts/preprocess.joblib - preprocessing pipeline.
- ? backend/app/member_1_attrition_risk/model_artifacts/schema.json - expected feature schema.
- ? backend/app/member_1_attrition_risk/model_artifacts/config.json - model config and threshold.

Frontend:

- ? ta/src/members/member_1_attrition_risk/RecruitmentAnalyticsPage.jsx - standalone risk scoring UI.
- ? ta/src/members/member_1_attrition_risk/HrCandidateReviewPage.jsx - HR candidate review and risk panel.
- ? ta/src/members/member_1_attrition_risk/README.md - member frontend notes.

APIs:

- ? GET /api/v1/attrition/schema
- ? POST /api/v1/attrition/score
- ? POST /api/v1/attrition/score-from-cv

Member 2 - Resume Parser

Purpose:

- ? Extract CV text from PDF/DOCX/TXT.
- ? Parse candidate profile details.
- ? Validate credentials.
- ? Produce resume explainability signals.

Backend:

- ? backend/app/member_2_resume_parser/routes.py - CV cache and text extraction API endpoints.
- ? backend/app/member_2_resume_parser/cv_parser.py - CV text extraction and feature inference.
- ? backend/app/member_2_resume_parser/resume_explainer_model.py - trained resume label/explainability model wrapper.
- ? backend/app/member_2_resume_parser/credential_validator.py - credential validation logic.
- ? backend/app/member_2_resume_parser/model_artifacts/resume_explainer.joblib - resume explainer artifact.
- ? backend/app/member_2_resume_parser/model_artifacts/resume_explainer_config.json - resume explainer config.

Frontend:

- ? ta/src/members/member_2_resume_parser/CandidateWorkspacePage.jsx - candidate portal, jobs, CV upload, applications.
- ? ta/src/members/member_2_resume_parser/ResumeParserPage.jsx - standalone resume parsing UI.

? ta/src/members/member_2_resume_parser/README.md - member frontend notes.

APIs:

? POST /api/v1/cv-cache/upload
? GET /api/v1/cv-cache/{cv_cache_id}/text

Member 3 - Job Matching

Purpose:

? Rank jobs for a CV.
? Rank candidates for a job.
? Show fit scores, matched terms, missing terms, and recommendations.

Backend:

? backend/app/member_3_job_matching/routes.py - standalone matching API endpoint.
? backend/app/member_3_job_matching/matching_engine.py - explainable matching/ranking engine.

Frontend:

? ta/src/members/member_3_job_matching/HrWorkspacePage.jsx - main recruiter workspace.
? ta/src/members/member_3_job_matching/CandidateMatchingPage.jsx - standalone matching UI.
? ta/src/members/member_3_job_matching/HrDashboardPage.jsx - HR dashboard.
? ta/src/members/member_3_job_matching/HrCandidatesPage.jsx - HR candidate list.
? ta/src/members/member_3_job_matching/HrPostJobsPage.jsx - job posting page.
? ta/src/members/member_3_job_matching/FullHiringCyclePage.jsx - full workflow demo page.
? ta/src/members/member_3_job_matching/README.md - member frontend notes.

APIs:

? POST /api/v1/matching/recommend

Member 4 - Interview Evaluation

Purpose:

? Score interview answers.
? Return communication/clarity/confidence style breakdowns.
? Produce a final band and recommendation summary.

Backend:

? backend/app/member_4_interview_evaluation/routes.py - interview evaluation API endpoint.
? backend/app/member_4_interview_evaluation/interview_scorer.py - model/rubric interview scoring logic.
? backend/app/member_4_interview_evaluation/model_artifacts/interview_scorer.joblib - interview scoring artifact.
? backend/app/member_4_interview_evaluation/model_artifacts/config.json - interview model config.

Frontend:

? ta/src/members/member_4_interview_evaluation/InterviewSoftSkillPage.jsx - standalone interview evaluation UI.
? ta/src/members/member_4_interview_evaluation/README.md - member frontend notes.

APIs:

? POST /api/v1/interview/evaluate

Shared Backend APIs

All routes below are registered under /api/v1 from backend/app/shared/hiring_cycle.py.

Auth:

? POST /api/v1/auth/register
? POST /api/v1/auth/login
? GET /api/v1/auth/me

Admin:

? POST /api/v1/admin/companies

- ? GET /api/v1/admin/companies
- ? GET /api/v1/admin/users
- ? GET /api/v1/admin/reports
- ? PATCH /api/v1/admin/users/{user_id}/access
- ? GET /api/v1/admin/ai-modules

Vacancies and public jobs:

- ? POST /api/v1/vacancies
- ? GET /api/v1/vacancies
- ? GET /api/v1/vacancies/{vacancy_id}
- ? PATCH /api/v1/vacancies/{vacancy_id}
- ? GET /api/v1/public/jobs
- ? POST /api/v1/public/jobs
- ? POST /api/v1/public/jobs/{vacancy_id}/evaluate-cv
- ? GET /api/v1/recruiter/vacancies

Candidate applications:

- ? POST /api/v1/vacancies/{vacancy_id}/apply
- ? GET /api/v1/candidates/applications

Recruiter workflow:

- ? GET /api/v1/recruiter/applications
- ? GET /api/v1/recruiter/candidates/compare
- ? POST /api/v1/recruiter/applications/{application_id}/recheck
- ? PATCH /api/v1/applications/{application_id}/status
- ? POST /api/v1/applications/{application_id}/decision
- ? GET /api/v1/recruiter/dashboard

Interview workflow:

- ? POST /api/v1/applications/{application_id}/interviews
- ? GET /api/v1/applications/{application_id}/interviews
- ? POST /api/v1/interviews/{interview_id}/submit

Other:

- ? GET /api/v1/status-catalog
- ? POST /api/v1/seed/info
- ? GET /health

Frontend Structure

Frontend entrypoints:

- ? ta/src/main.jsx - React root, providers, and router setup.
- ? ta/src/App.jsx - route definitions and protected candidate/HR routing.
- ? ta/src/index.css - global styles.
- ? ta/src/App.css - app-level styles.

Shared frontend layer:

- ? ta/src/shared/components/ - shared UI components.
- ? ta/src/shared/components/portal/WorkspaceShell.jsx - shared candidate/HR workspace shell.
- ? ta/src/shared/context/AuthSessionContext.jsx - login/session context.
- ? ta/src/shared/context/SharedCvContext.jsx - shared CV upload/cache state.
- ? ta/src/shared/context/ThemeContext.jsx - light/dark theme context.
- ? ta/src/shared/context/UserModeContext.jsx - candidate/HR mode context.
- ? ta/src/shared/data/landingContent.js - landing content data.
- ? ta/src/shared/pages/PortalLoginPage.jsx - candidate/HR login page.
- ? ta/src/shared/pages/NotFoundPage.jsx - fallback route.
- ? ta/src/shared/pages/LandingPage.jsx - legacy/shared landing page.
- ? ta/src/shared/pages/ModulePlaceholderPage.jsx - placeholder page.
- ? ta/src/shared/utils/portalApi.js - auth/API helper.
- ? ta/src/shared/utils/hrWorkspaceApi.js - HR API helper.
- ? ta/src/shared/utils/modelUtils.js - JSON/export helpers.
- ? ta/src/shared/utils/motion.js - animation variants.

Frontend routes:

- ? / - redirects to /candidate
- ? /candidate - candidate login
- ? /candidate/dashboard - candidate dashboard section
- ? /candidate/jobs - candidate jobs section
- ? /candidate/jobs/:jobId - candidate job detail/apply flow
- ? /candidate/applications - candidate applications section
- ? /hr - HR login
- ? /hr/dashboard - HR dashboard section
- ? /hr/jobs - HR job posting/listing section
- ? /hr/candidates - HR candidate list section
- ? /hr/candidates/:applicationId - HR candidate review page
- ? /hr/interviews - HR interview evaluation section
- ? /hr/analytics - HR analytics section
- ? * - not found page

Shared Components

Key shared UI files:

- ? AboutProject.jsx
- ? ContactSection.jsx
- ? FeatureCard.jsx
- ? FeatureCards.jsx
- ? Footer.jsx
- ? Header.jsx
- ? Hero.jsx
- ? HowItWorks.jsx
- ? MinimalInputCard.jsx
- ? MinimalOutputCard.jsx
- ? ModulePageLayout.jsx
- ? ModulesGrid.jsx
- ? Overview.jsx
- ? SectionTitle.jsx
- ? TechStack.jsx
- ? ThemeToggleButton.jsx
- ? portal/WorkspaceShell.jsx

Build And Verification

Backend syntax/import check:

```
cd backend
source .venv/bin/activate
python -m compileall app
python -c "from app.main import app; print(app.title); print(len(app.routes))"
```

Frontend build:

```
cd ta
npm run build
```

Ownership Summary

- ? Member 1 owns attrition risk backend and frontend.
- ? Member 2 owns resume parsing backend and frontend.
- ? Member 3 owns job matching backend and recruiter workflow frontend.
- ? Member 4 owns interview evaluation backend and frontend.
- ? Shared backend/frontend folders contain integration code needed to combine all four member components into one working application.